A. Completed evidence

every result reported in this paper

Unmodified firmware logic
smart_home.ino compiled against host stubs
for Arduino, FreeRTOS, WiFi, MQTT, NVS, OLED.
Not altered for testing.

Campaign (E1, E2)
6,000 trials, seed 1000
rates 0 to 20,000 msg/s
fork() per trial

Ablation (E3)
3,600 trials, seed 2000
12 configurations
300 trials each

Fuzzer (E4)
3,000,000 runs
seed 7, 6 input classes
ASan + UBSan

Broker testbed (E6)
real Mosquitto 2.0.18
TLS, per-client ACLs
8 scenarios

Sensitivity sweep (E5) and corpus study
180 grid cells; 135 projects, commit-pinned,
guard rule with 8 controls, 0 hits

Limitation that defines the open step
The harness advances a logical clock. No radio,
no parallelism, no contention: zero variance.

B. Not yet measured

rig specified; no result in this paper comes from it

Device under test
ESP32 DevKit
running v2.2.0
unmodified

GPIO

Measuring board
second ESP32,
independent clock,
logic probe

Four conditions

- **A.** idle (control)
- **B.** legitimate network load
- **C.** command-interface stress
- **D.** radio-layer stress

A to C need no RF isolation.
D: authorised, isolated RF environment only.

Dependent variable
Physical sensor-to-siren latency distribution,
compared across conditions with analyse_hw.py.

Conditions A to C need two boards and a probe.
No attack tooling is published.